

Quantifying the environmental cost of everyday AI use, looking for seeking scientific feedback on a personal impact tool

Posted by u/slimhassoonny • 0 points • 0 comments

Hi everyone,

I've been researching the environmental costs of large AI models (specifically Large Language Models) and how they compare to more familiar carbon-emitting activities. While we often focus on transportation, agriculture, or manufacturing, I think the digital side of emissions (especially from fast-growing technologies like AI) is under-discussed.

A few references that caught my attention:

- * One AI-generated image can emit as much CO₂ as driving ~3 miles in a typical gas-powered vehicle. [Source](<https://arxiv.org/abs/2311.16863>)
- * Training GPT-3 emitted an estimated 552 metric tons of CO₂, roughly equivalent to 500 round-trip flights from New York to San Francisco. [Source] (<https://news.mit.edu/2025/explained-generative-ai-environmental-impact-0117>)
- * Even a single ChatGPT query consumes significantly more energy than a Google search—about 5× more.
- * These systems also consume notable amounts of water, with inference-related water usage reaching ~500 mL per conversation in some data centers. [Source] (<https://www.washingtonpost.com/technology/2024/09/18/energy-ai-use-electricity-water-data-centers/>)

I'm currently prototyping a browser extension to help users visualize the digital footprint of their AI interactions. The goal is not to shame use, but to provide:

1. A real-time footprint score (CO₂ + water estimate) after each ChatGPT session
2. A basic tracker to show trends over time
3. Small behavioural suggestions to lower impact (e.g., using more concise queries or less resource-intensive models, maybe pushing for Google searches depending on the query)

I'm not trying to promote a product here, just looking to get early scientific feedback from a community that takes climate data seriously.

- * Does this kind of tool have scientific value for raising awareness?
- * What pitfalls should I avoid when estimating digital emissions in real time?
- * Any important peer-reviewed work I should include in my methodology?

If interested, here's the prototype page:

 <https://gaiafootprint.carrd.co>

Thank you!

Comments